Aga Khan University Examination Board

Notes from E-Marking Centre on SSC-II Mathematics Examination May 2018

Introduction:

This document has been produced for the teachers and candidates of Secondary School Certificate (SSC-II) Mathematics. It contains comments on candidates' responses to the 2018 SSC-II Examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes:

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that require candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations:

Generally it is noted that weaker candidates are not well-versed with the hierarchy of arithmetical, algebraic operations, appropriate formulae and its application. This is generally obstructing their performance in overall paper of mathematics.

Detailed Comments:

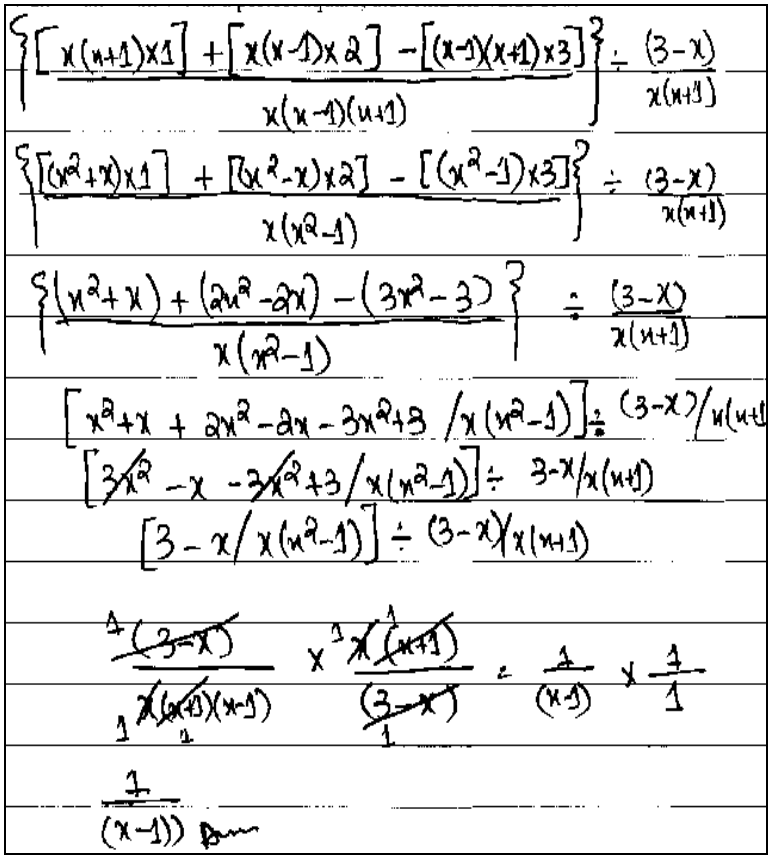
Constructed Response Questions (CRQs)

Question 1a:

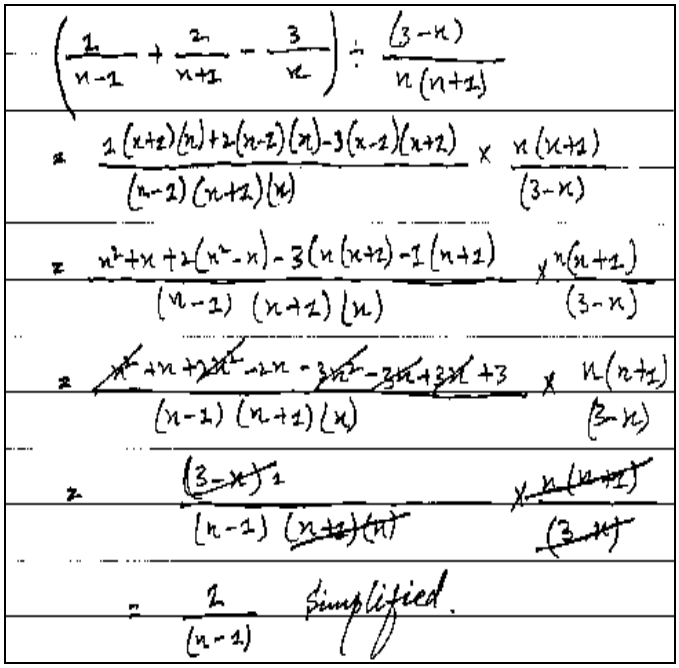
Simplify completely $\left(\frac{1}{x-1}+\frac{2}{x+1}-\frac{3}{x}\right)\div\frac{(3-x)}{x(x+1)}$.

Better responses indicated that candidates had command over the concept of LCM. Candidates correctly found the LCM and wrote numerator accordingly. Finally, they simplified the given algebraic expressions using correct hierarchy of algebraic operations.

Example 1:



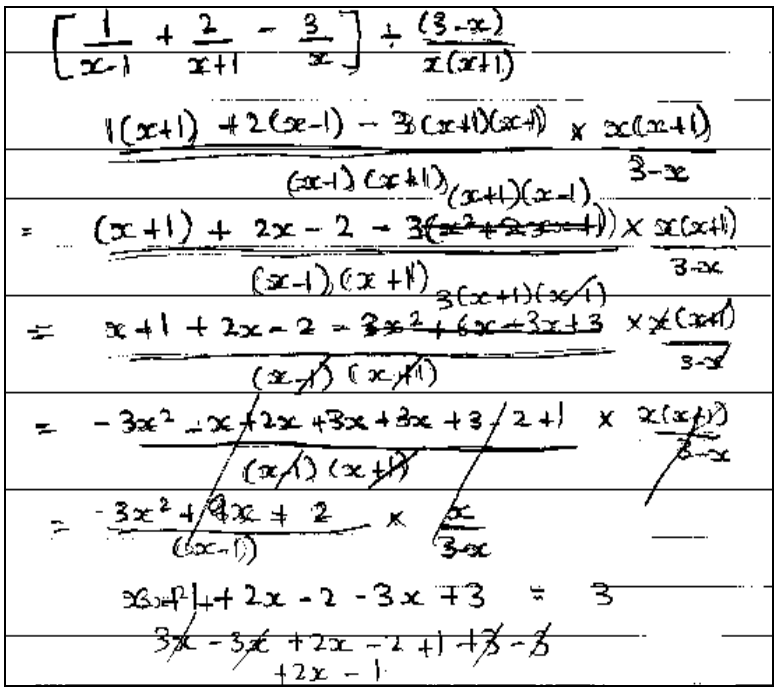
Example 2:



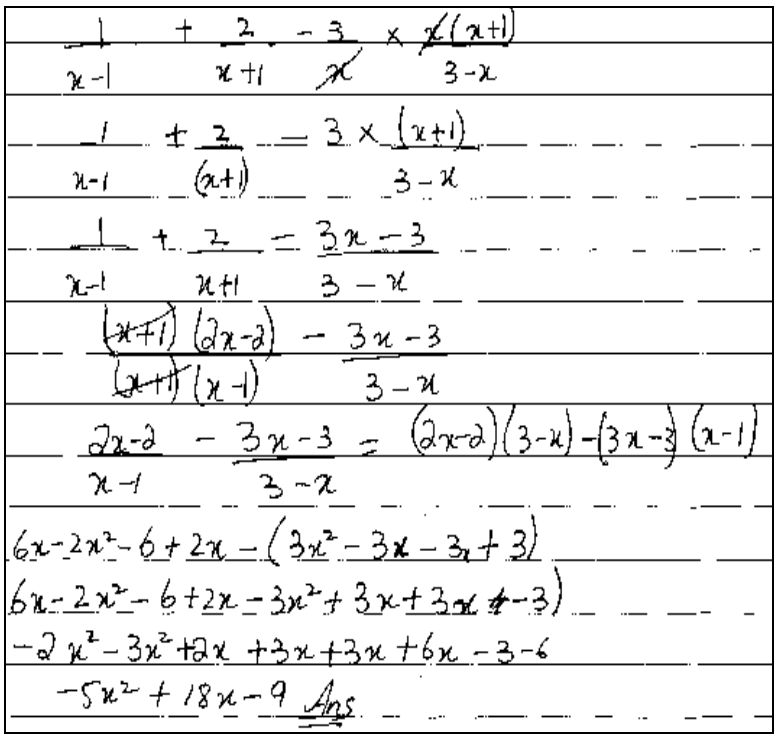
Weaker responses showed that candidates have lack of understanding of the concept of LCM. They also made mistakes in applying arithmetic operations to simplify the given algebraic expressions and consequently failed to simplify the given algebraic expression correctly.

In few other responses, it was noted that candidates took $(x-1)(x+1)$ instead of $x(x-1)(x+1)$ and made mistakes in the process of application of arithmetic operations on algebraic expression as well.

Example 1:



Example 2

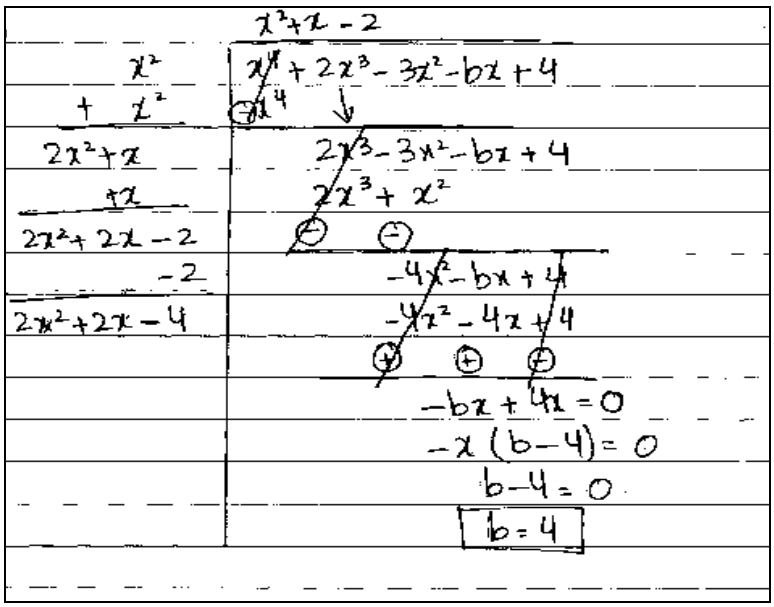


Question 1b:

If $x^4 + 2x^3 - 3x^2 - bx + 4$ is a perfect square, then find the value of *b*.

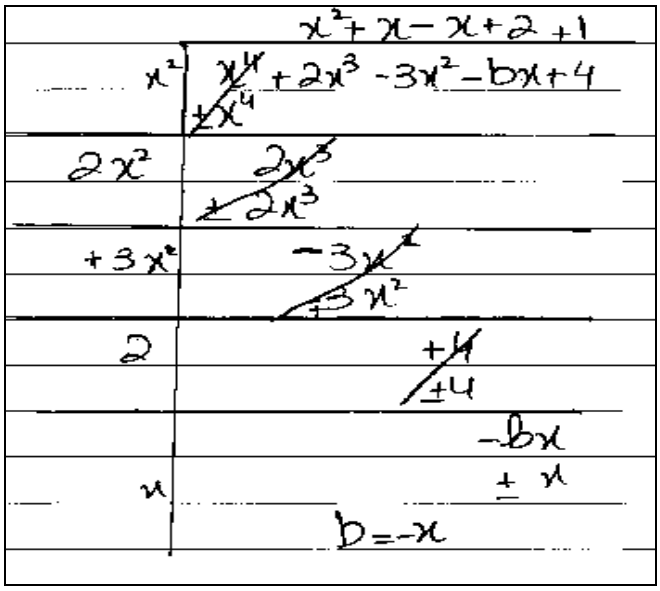
Better responses displayed that candidates comprehended the question well and applied the correct process of division method to find the square root of the given algebraic expression and compared the remainder with zero, i.e. $-bx + 4x = 0$ to find the value of *b*.

Example:

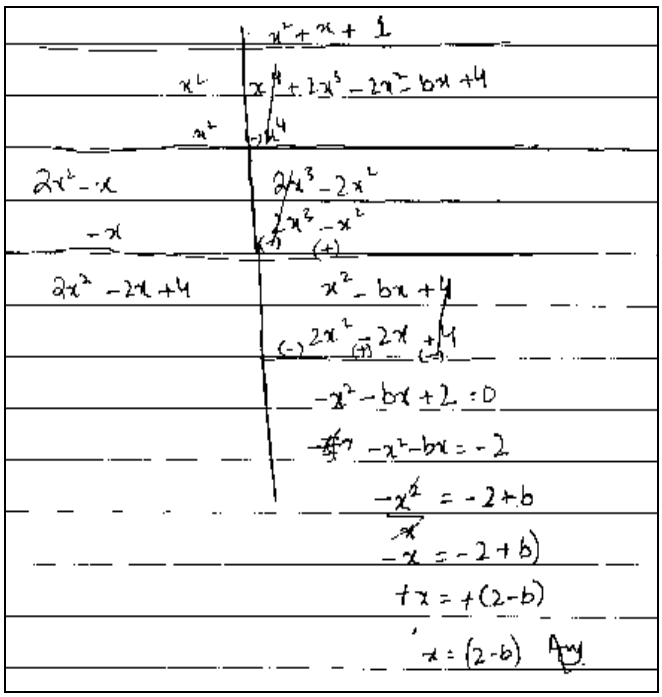


Weaker responses reflected that candidates made mistakes in the division process. They failed to follow the steps needed to find the square root of the given algebraic expression and consequently, were unable to find the value of b. The mistakes are noted in writing divisor, quotient and sign in the process of addition and subtraction of algebraic terms. It is also observed that weaker candidates were failed to compare the coefficient of *x* properly, as evident from the given examples.

Example 1:



Example 2:

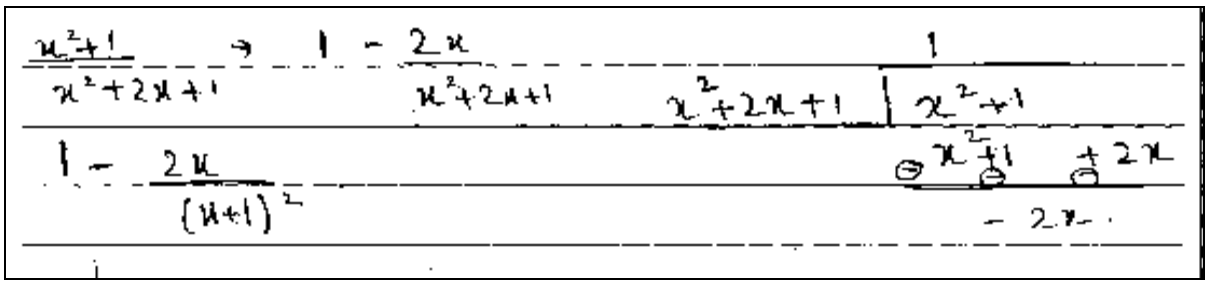


Question 2a:

Convert $\dfrac{x^2+1}{x^2+2x+1}$ into proper fraction.

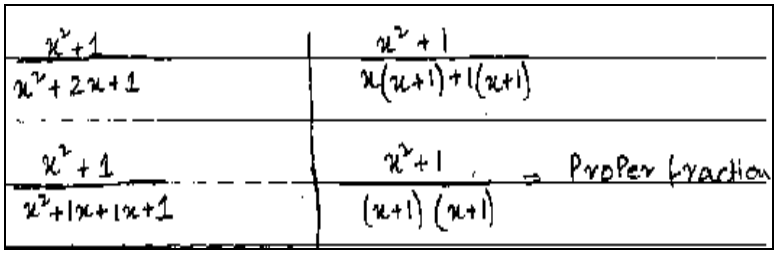
Better responses exhibited that the candidates correctly divide the numerator by denominator and converted the given expression into proper fraction.

Example:

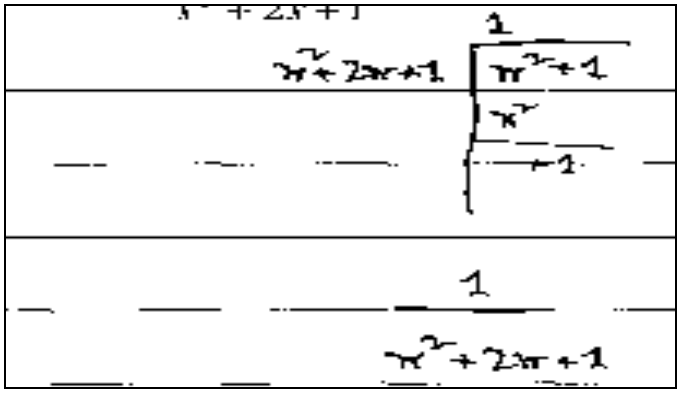


Weaker responses revealed that candidates made different types of mistakes, which included incorrect division process, mistakes in writing the signs of algebraic terms while adding and subtracting these terms. In a few other responses, it was noted that candidates having no concepts of proper fraction.

Example 1:



Example 2:

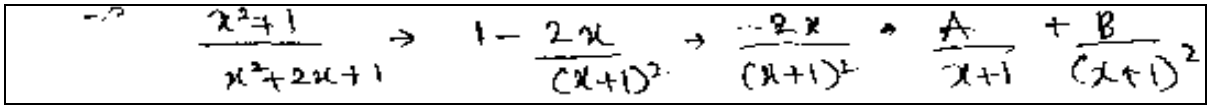


Question 2b:

Write correct form required to resolve $\frac{x^2+1}{x^2+2x+1}$ into its partial fractions.

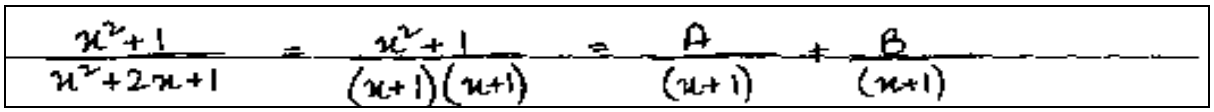
Better responses exhibited that the candidates wrote the required form correctly, i.e. denominator contains linear repeated factors.

Example:



Weaker responses revealed that candidates were unable to pick the correct form and failed to respond to the question correctly.

Example:



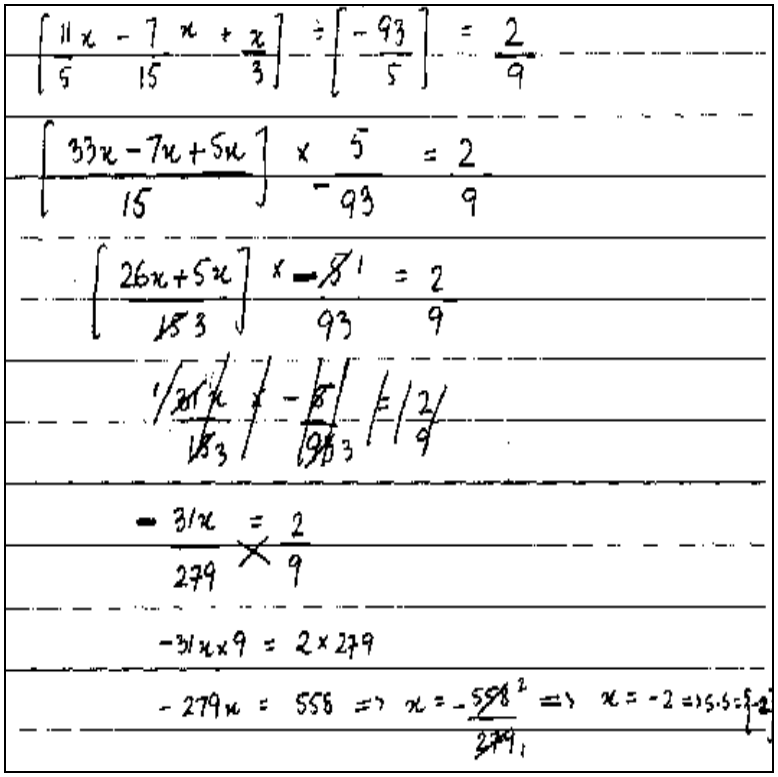
Question 3a:

Find the solution set of the linear equation $\left(2\frac{1}{5}x-\frac{7}{15}x+\frac{x}{3}\right)\div\left(-18\frac{3}{5}\right)=\frac{2}{9}$.

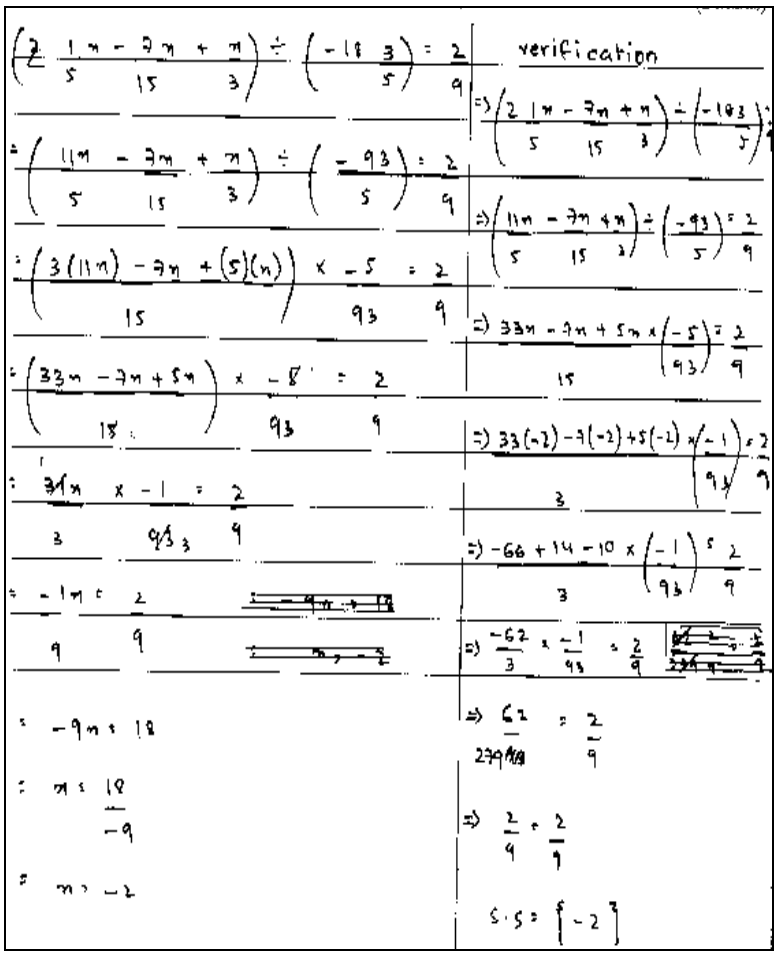
Generally, it was a well-attempted question and most candidates chose to attempt it.

Better responses displayed that candidates systematically solved the given linear equation by using correct mathematical steps. In a few cases, candidates simplified $\left(2\frac{1}{5}x-\frac{7}{15}x+\frac{x}{3}\right)$ first by taking LCM then converted $-18\frac{3}{5}$ into $-\frac{93}{5}$. After that they collected the terms containing *x* on one side and constant terms on other sides. In a few other responses, it was noted that the candidates took LCM on the LHS and multiplied both sides by the LCM and performed simplification process systematically to get the value of *x.* Although, solution set and verification was not asked in the question but few candidates also wrote the solution set and done the process of verification.

Example 1:

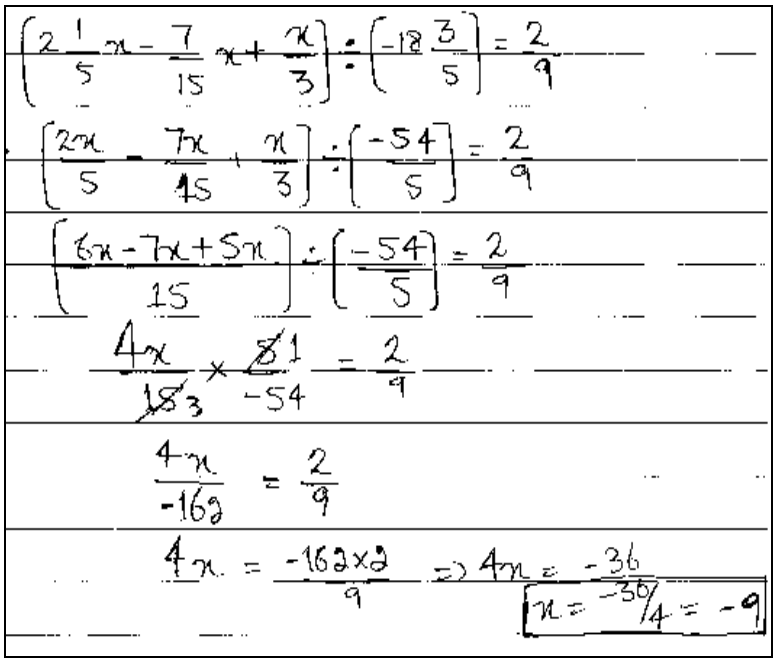


Example 2:

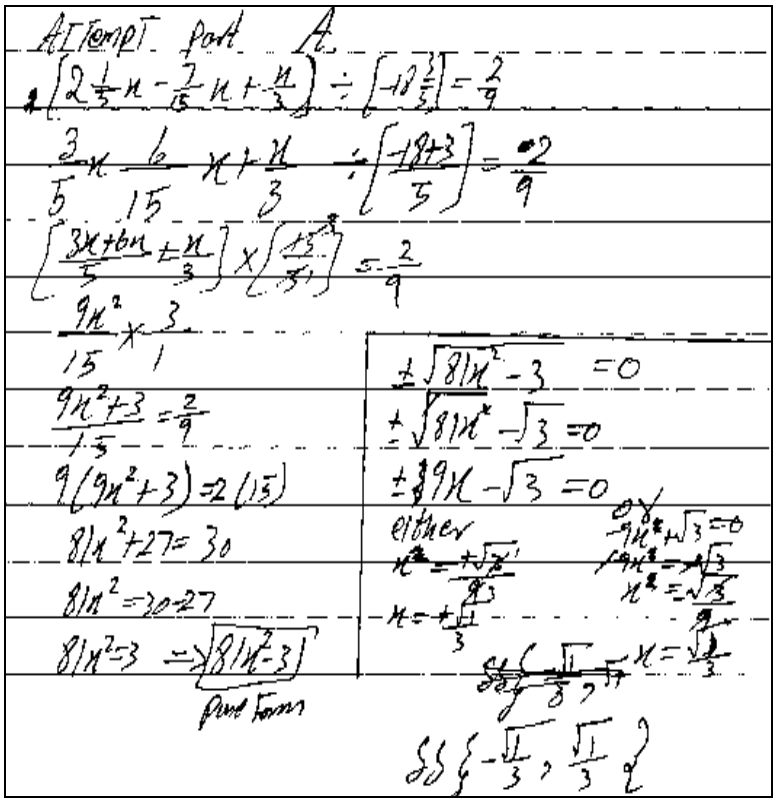


Weaker responses exhibited that candidates failed to solve the given linear equation. In some responses, the candidates failed to perform the algebraic operation correctly on the given equation. Few other mistakes have been presented in the following examples.

Example 1:



Example 2:



Question 3b:

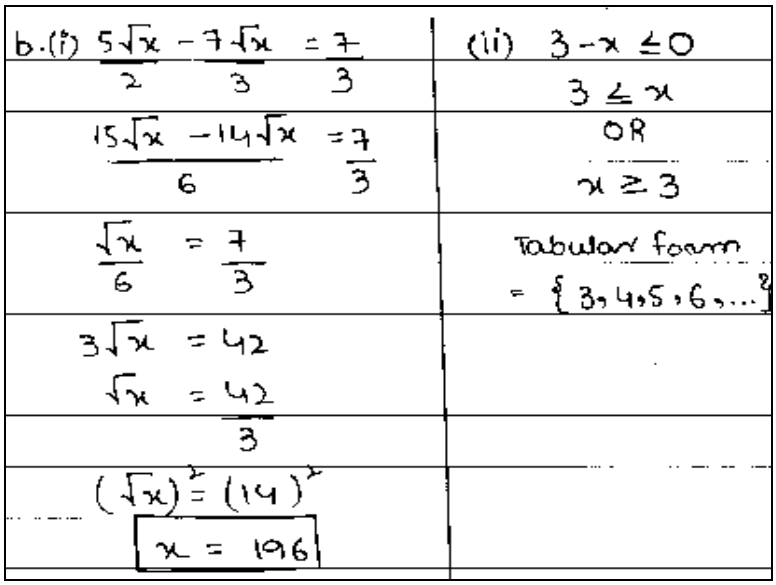
i. Find the value of x for the equation $\frac{5\sqrt{x}}{2} - \frac{7\sqrt{x}}{3} = \frac{7}{3}$.

(**Note**: Verification is not required)

ii. Find the solution set of $3 - x \leq 0$ in tabular form, where $x \in W$.

Better responses showed that candidates correctly applied the steps to solve the equation involving square roots or radical sign to find the solution set of the given equation. They rightly took the square after simplification to get the value of *x*. In part **ii.** candidates were able to solve the inequality and wrote the answer in tabular form to fulfill the requirement of the question.

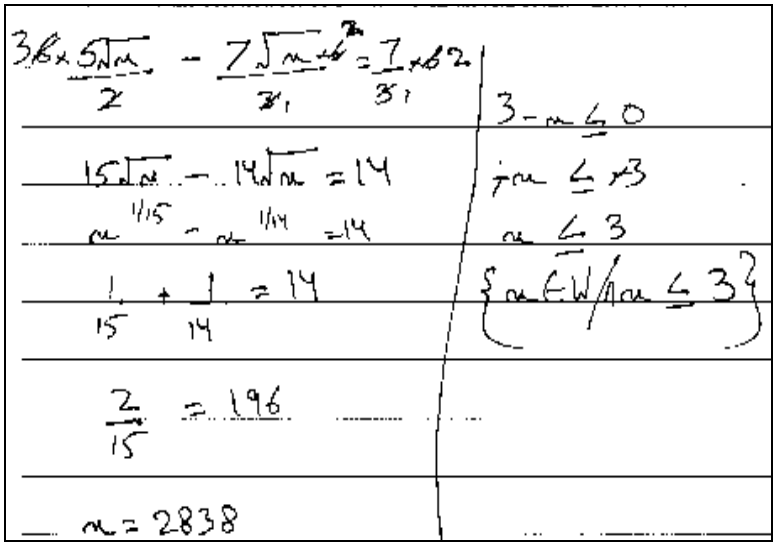
Example:



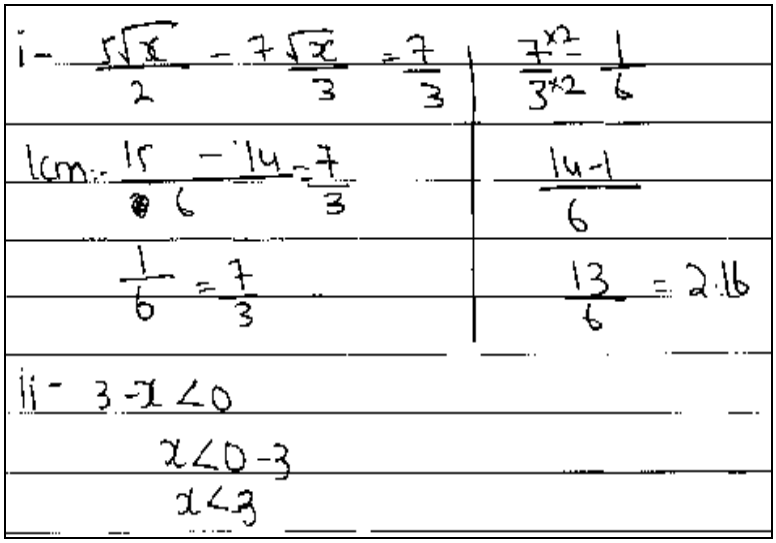
Weaker responses displayed that candidates were confused in solving equation involving radical sign and inequalities. The weaker responses also showed mistakes in performing algebraic operations.

Few other mistakes have been noted in the examples cited below.

Example 1:



Example 2:



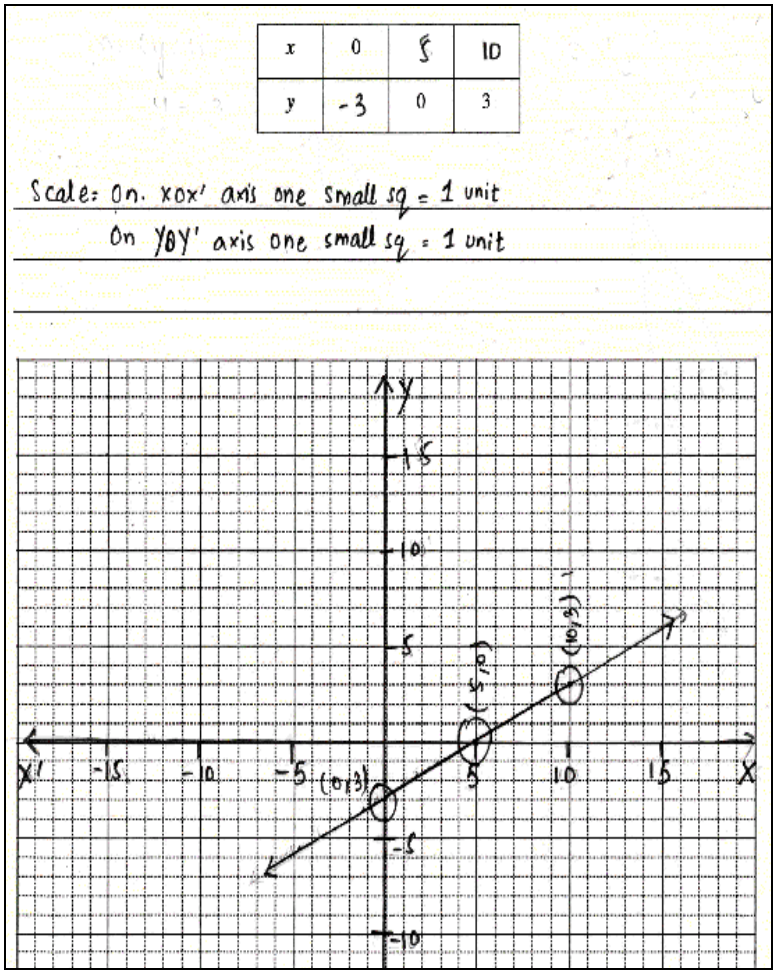
Question 4:

Complete the following table for the equation $3x-5y=15$ and draw its graph on the given graph.

x	0		
y		0	3

Better responses displayed that candidates correctly found the unknown values of *x* and *y* by substituting values in the given equation to complete the given table. Candidates skillfully drew the given linear equation on the given graph paper by selecting appropriate scales on *x*-axis and *y*-axis.

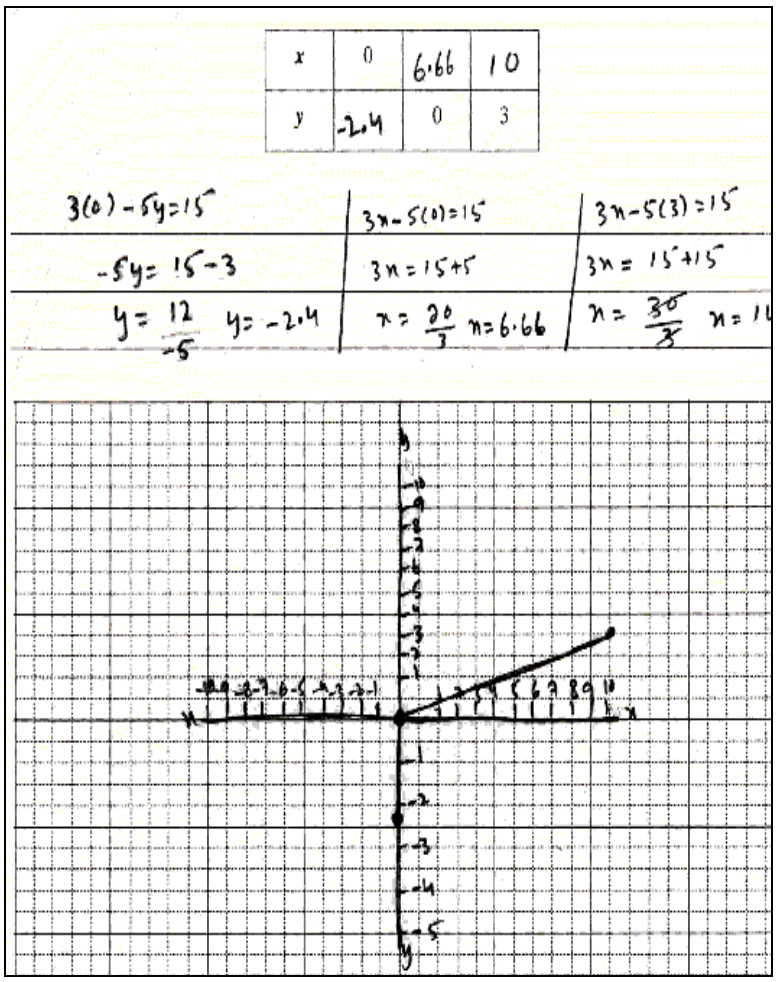
Example:



Weaker responses displayed various types of mistakes. Some common mistakes have been listed below.

- Candidates failed to find the unknown values of *x* and *y*, which is indicative of the fact that the candidates are weak in applying basic arithmetic operations to solve linear equation in two variables.
- Candidates failed to take the proper scale on x-axis or y-axis
- Candidates showed misconceptions in locating points on the graph papers. They interchanged the values of *x* and *y* and failed to differentiate between the – and + sign while locating the points on the graph paper.
- Other misconceptions in drawing graph can be seen in the following examples.

Example 1:



x	0	6.66	10
y	-2.4	0	3

$3(0) - 5y = 15$ | $3x - 5(0) = 15$ | $3x - 5(3) = 15$

$-5y = 15 - 3$ | $3x = 15 + 5$ | $3x = 15 + 15$

$y = \frac{12}{-5}$ $y = -2.4$ | $x = \frac{20}{3}$ $x = 6.66$ | $x = \frac{30}{3}$ $x = 10$

Example 2:



x	0	2	0
y	1	0	3

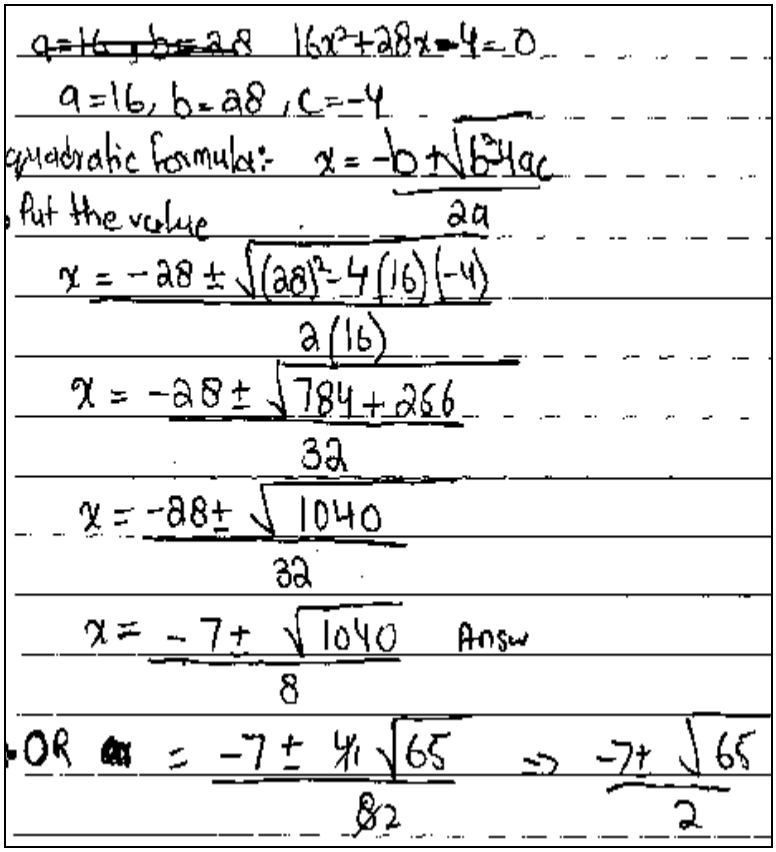
Question 5:

Use quadratic formula to find the values of x for $16x^2 + 28x = 4$.

It was generally well-attempted question.

Better responses indicated that candidates were well-versed with the use of quadratic formula. They correctly identified *a*, *b* and *c* and substituted in the quadratic formula to find the values of *x*.

Example 1:



Weaker responses indicated that candidates failed to follow the proper steps needed to solve the given quadratic equation. They made mistakes at various stages. Listed below are some common mistakes.

$16x^2 + 28x = 4$ was written as $16x^2 + 28x + 4 = 0$ or $16x^2 + 28x + 4$

Few responses indicated that candidates made different mistakes in writing formula which are cited below.

$$x = -b \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

$$x = -b \pm \frac{\sqrt{b^2 + 4ac}}{2a}$$

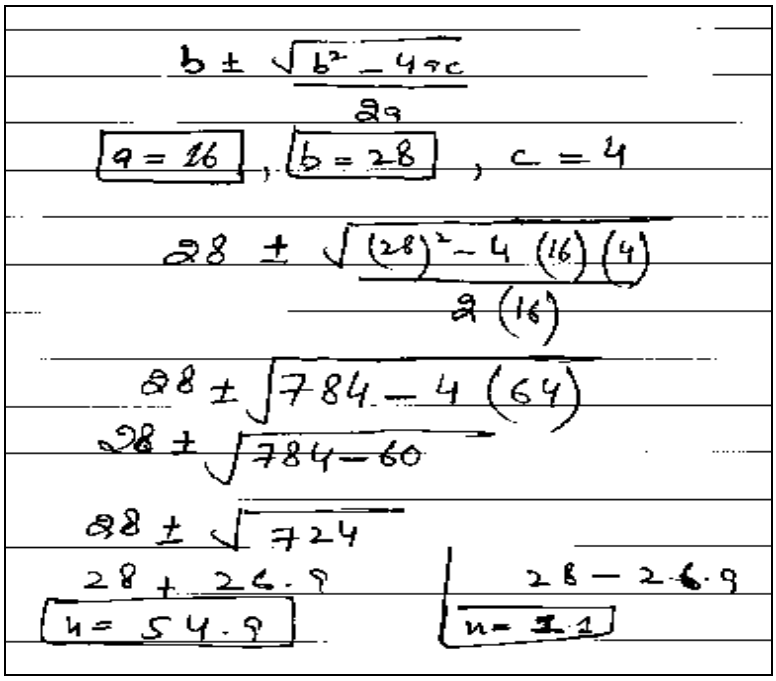
$$x = \frac{-b \pm \sqrt{b^2 + 4ac}}{2a}$$

$$x = \frac{b \pm \sqrt{b^2 + 4ac}}{2a}$$

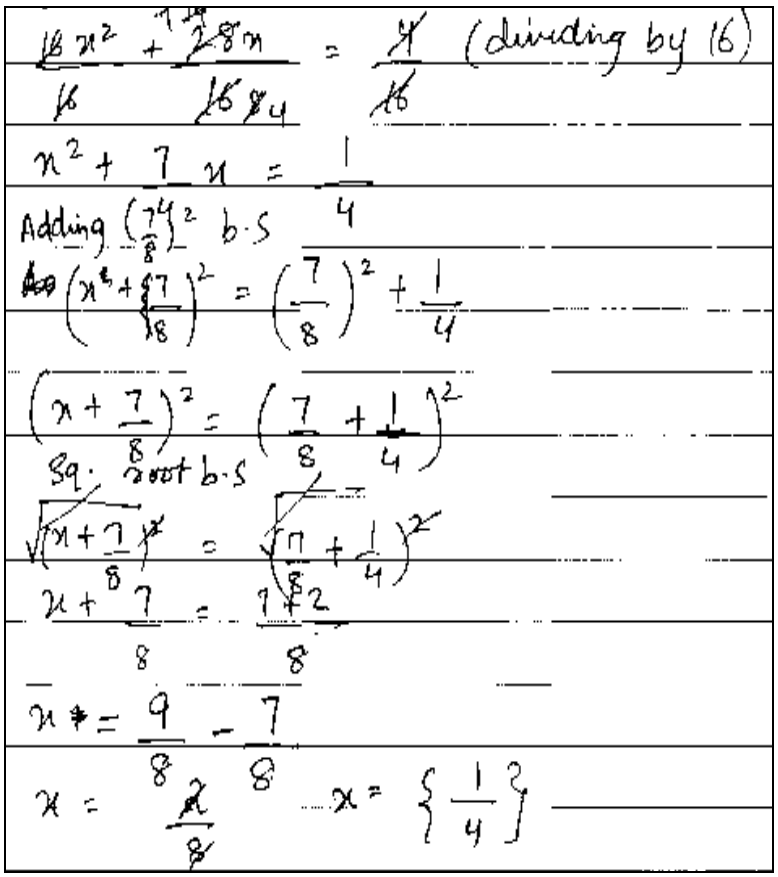
$$x = \frac{b \pm \sqrt{b^2 - 2ac}}{4a}$$,etc.

Therefore, candidates failed to find the values of x for the given quadratic equation. Few other mistakes are evident from the following examples.

Example 1:



Example 2:



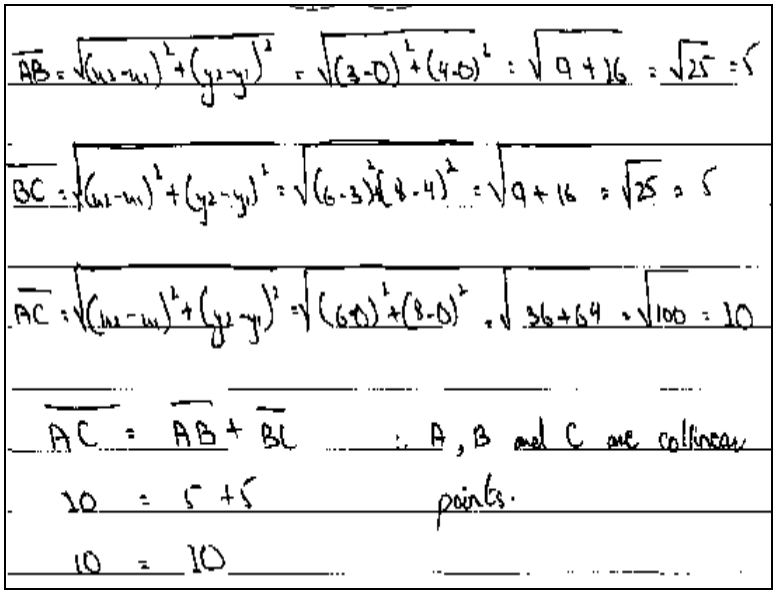
Question 6a:

Prove that the points *A* (0, 0), *B* (3, 4) and *C* (6, 8) are collinear points.

There was a choice between part **a** and **b** of the question. Most of the candidates opted for part **a.** Generally it was a well-attempted question.

Better responses exhibited that the candidates wrote the correct distance formula, substituted the values of the coordinates correctly and were able to prove that the give points *A* (0, 0), *B* (3, 4) and *C* (6, 8) are collinear. They showed that *AB +BC= AC.*

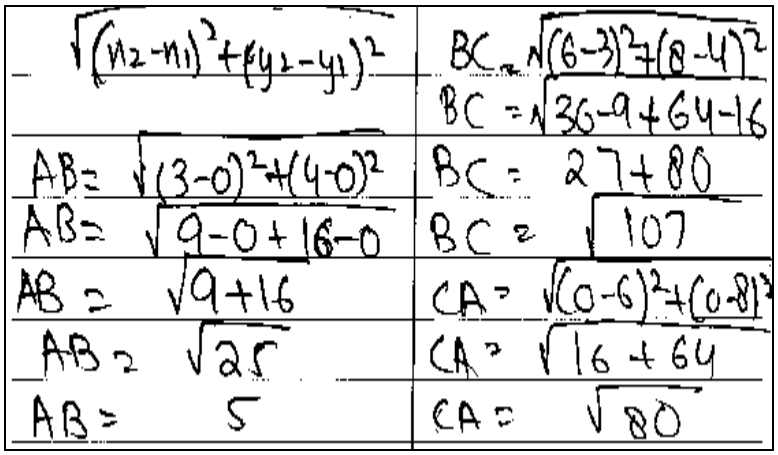
Example:



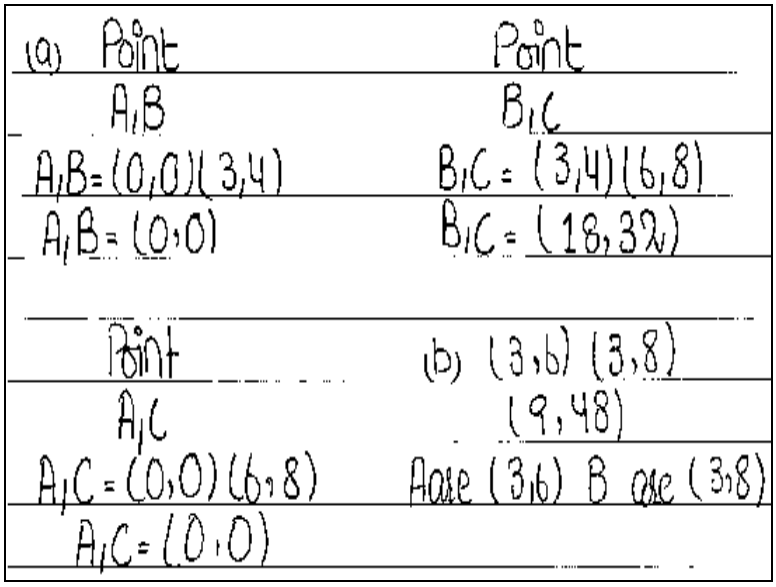
Weaker responses showed that candidates were unable to comprehend the question which resulted in different types of mistakes. Few of those mistakes are as follows:

- Wrote wrong formula for example, $|AB| = \sqrt{(x_2-x_1)+(y_2-y_1)}$,
  $|AB| = \sqrt{(x_2+x_1)^2-(y_2+y_1)^2}$, $|AB| = \sqrt{(y_2-x_1)^2+(x_2-y_1)^2}$ or
  $AB = (\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$
- Made mistakes in substitution of values of the coordinates in the formula.
- Made mistakes in simplification after substituting the values in the formula.
- Few other mistakes and conceptual errors are obvious in the given examples.

Example 1:

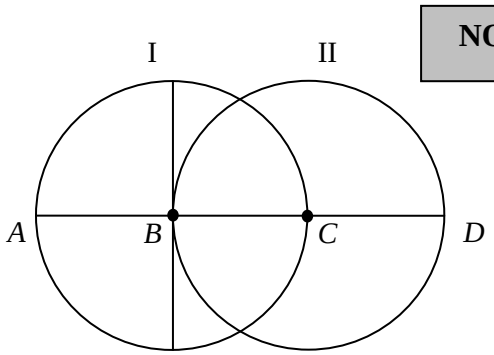


Example 2:



Question 6b:

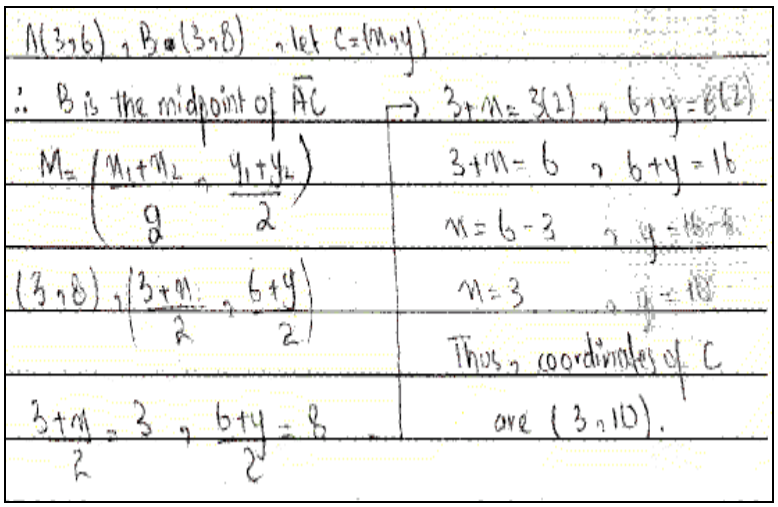
In the given diagram, two circles I and II have same radii. *B* is the centre of circle I and *C* is the centre of circle II. If the coordinates of *A* are (3, 6) and *B* are (3, 8), then find coordinates of the centre of circle II.



This question was based on the application of midpoint formula. Few candidates chose to attempt this question. However, the candidates who attempted this question generally performed well.

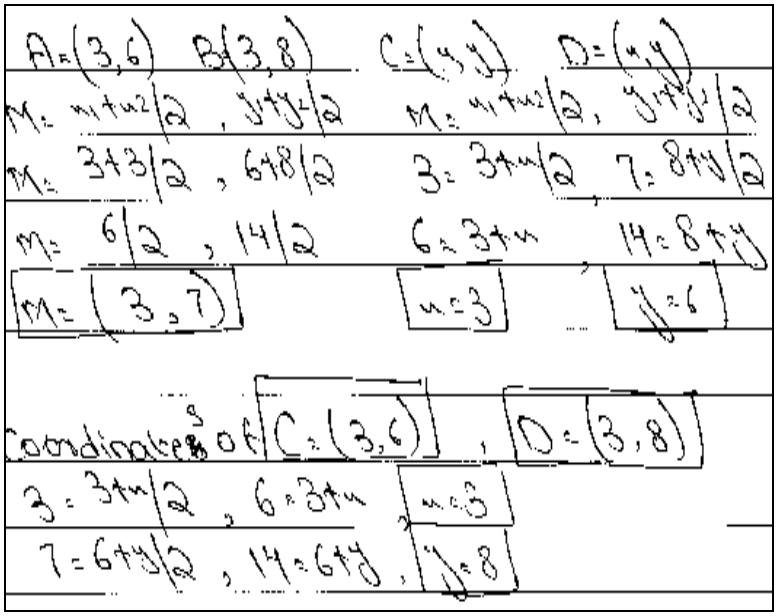
Better responses indicated that candidates comprehended the question well and used correct logic to find the coordinates of the centre of the circle II by using midpoint formula in correct order.

Example:



Weaker responses showed that candidates were failed to comprehend the question or applied the formula of midpoint inappropriately. Hence, they failed to find the coordinates of the centre of the circle II. In some responses, it was noted that candidates wrote incorrect formula to find the midpoint or applied the distance formula instead of midpoint formula.

Example 1:



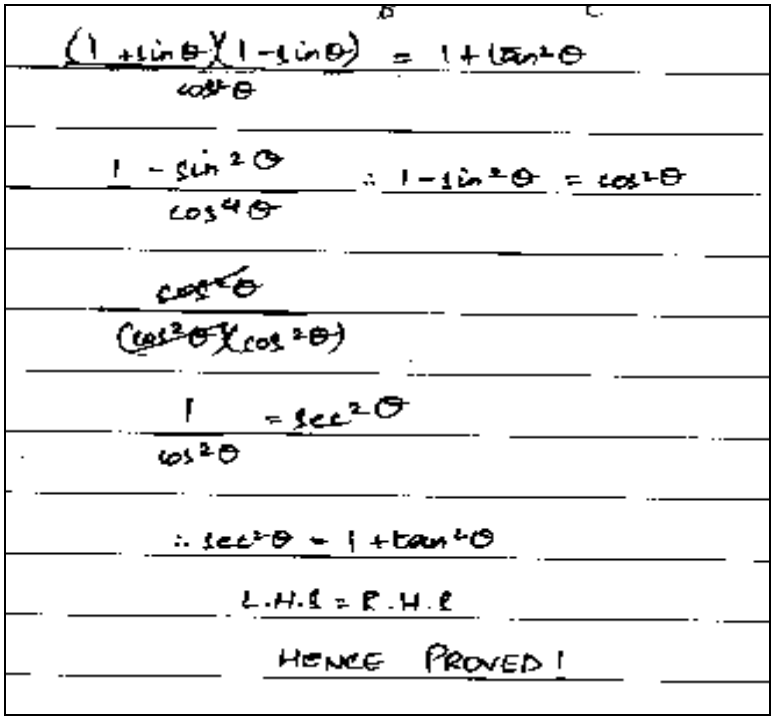
Question 7a:

Prove that $\dfrac{(1+\sin\theta)(1-\sin\theta)}{\cos^4\theta} = 1+\tan^2\theta$.

This question offered a choice between part **a** and part **b**. Candidates mostly opted to attempt part **a**. The question was based on the application of trigonometric identities to prove the required result.

Better responses showed that candidates aptly applied the formulae and trigonometric identities to prove the required result.

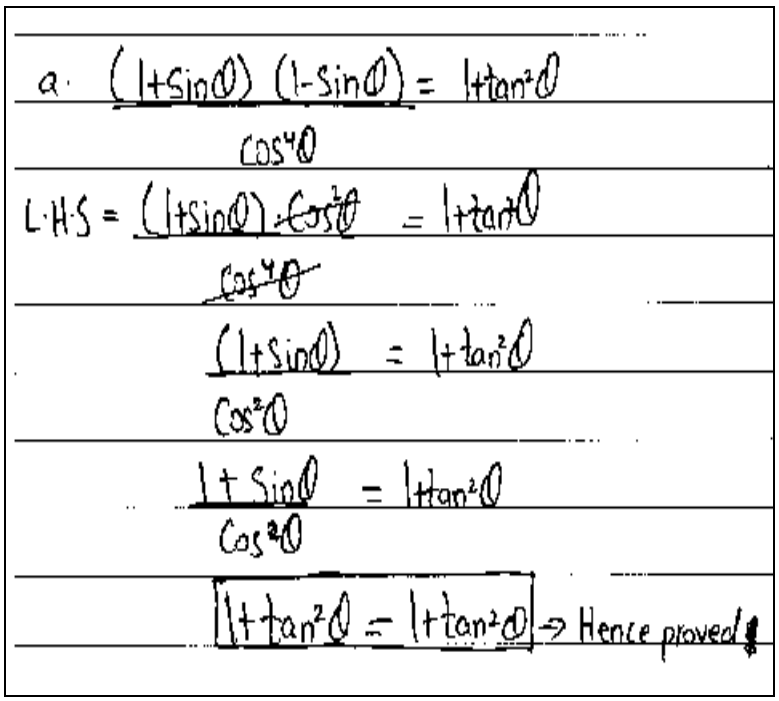
Example:



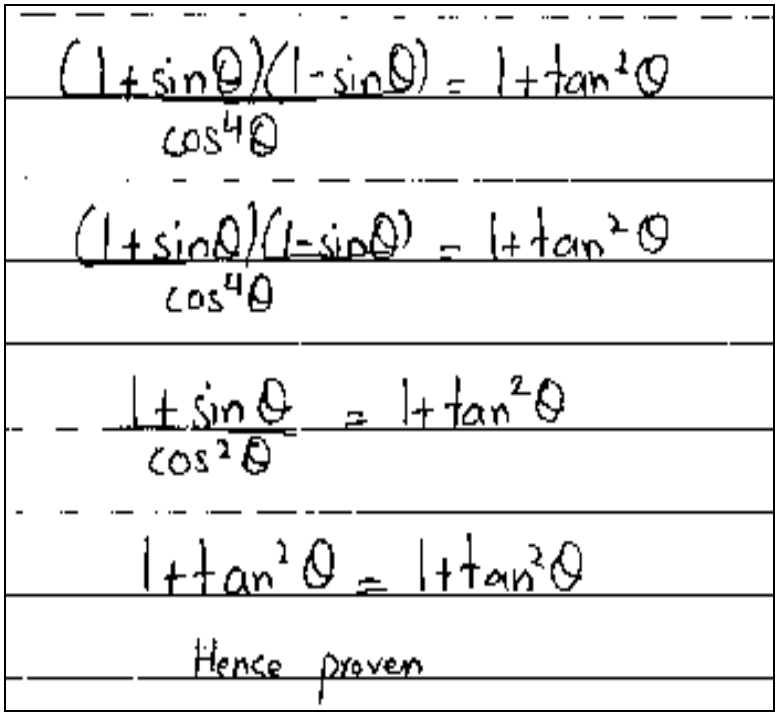
Weaker responses displayed that candidates made various mistakes. Few of them are listed below:

They wrote $1-\sin\theta = \cos^2\theta$ instead of $1-\sin^2\theta = \cos^2\theta$ and failed to exhibit use of correct identity $1+\tan^2\theta = \sec^2\theta$. Therefore, consequently failed to prove required result.

Example 1:



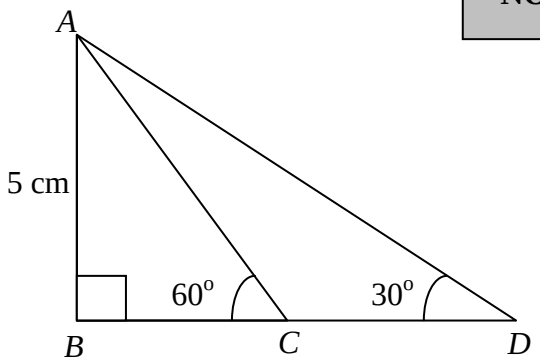
Example 2:



Question 7b:

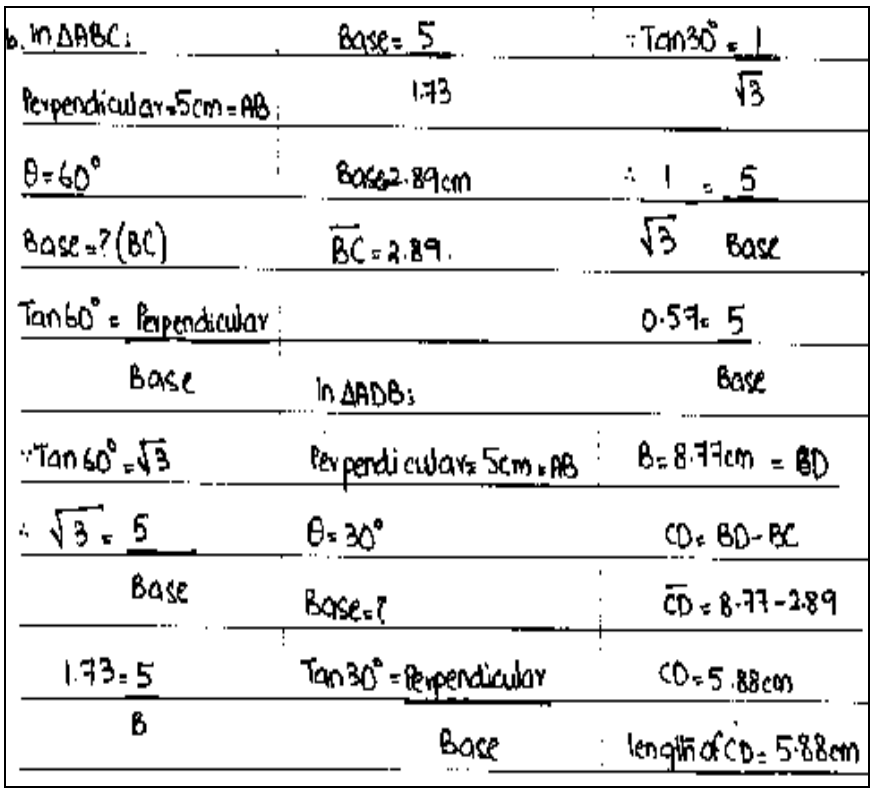
In the given diagram, find the length of *CD*.

NOT TO SCALE



Better responses showed that the candidates had good understanding of the trigonometric ratios and their application to find the unknown sides of the given triangle. They systematically found lengths of *BD* and *BC* and then, calculated the difference in the lengths of *BD* and *BC* to find *CD*.

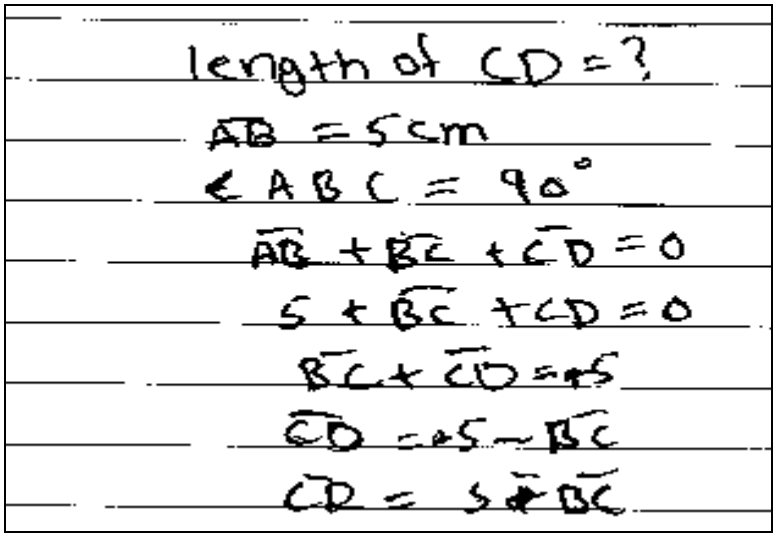
Example:


b. In ΔABC:
Perpendicular = 5cm = AB
θ = 60°
Base = ? (BC)
Tan 60° = Perpendicular / Base
∴ Tan 60° = √3
∴ √3 = 5 / Base
1.73 = 5 / B
Base = 5
1.73
Base = 2.89cm
$\overline{BC}$ = 2.89.
In ΔABD:
Perpendicular = 5cm = AB
θ = 30°
Base = ?
Tan 30° = Perpendicular / Base
∴ Tan 30° = 1 / √3
∴ 1 / √3 = 5 / Base
0.57 = 5 / Base
B = 8.77cm = BD
CD = BD − BC
$\overline{CD}$ = 8.77 − 2.89
CD = 5.88cm
length of CD = 5.88cm

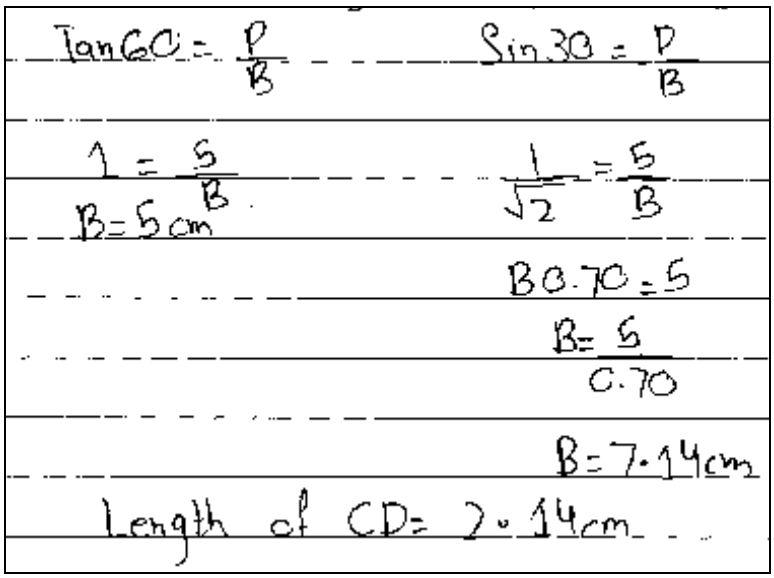
Weaker responses displayed lack of understanding of trigonometric ratios and their application to find unknown lengths and thus failed to find the length of *CD*. Few of the mistakes are given as follows:

They failed to apply the correct trigonometric ratio or they applied correct trigonometric ratios incorrectly. They also wrote wrong values of trigonometric ratios and therefore, failed to find the required length.

Example 1:

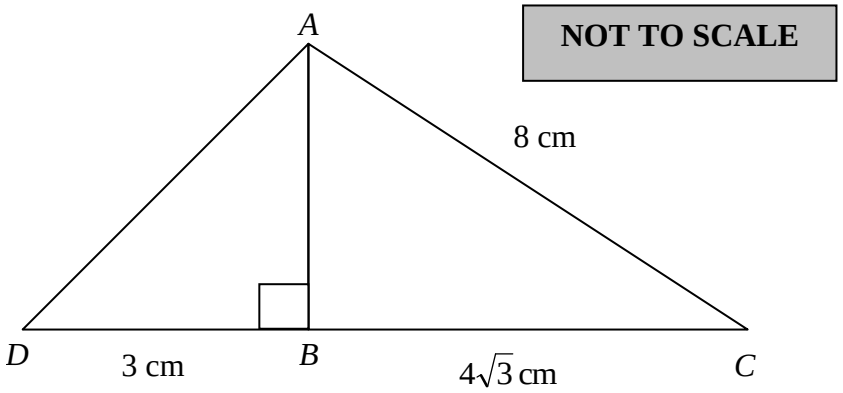


Example 2:



Question 8a:

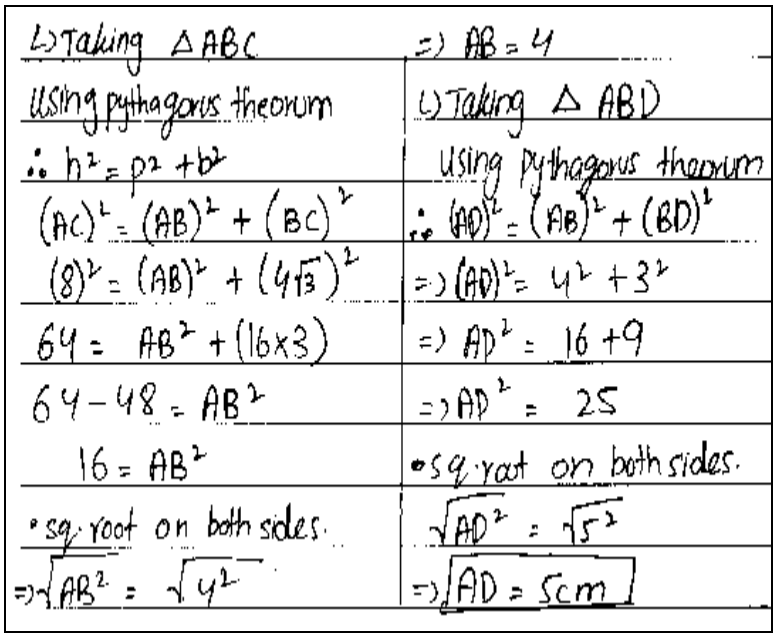
In the given diagram, find the length of *AD*.



The question was generally well-attempted. The question offered a choice between part **a** and part **b**. Candidates mostly attempted part **a** of the question which was based on Pythagorean Theorem and its application.

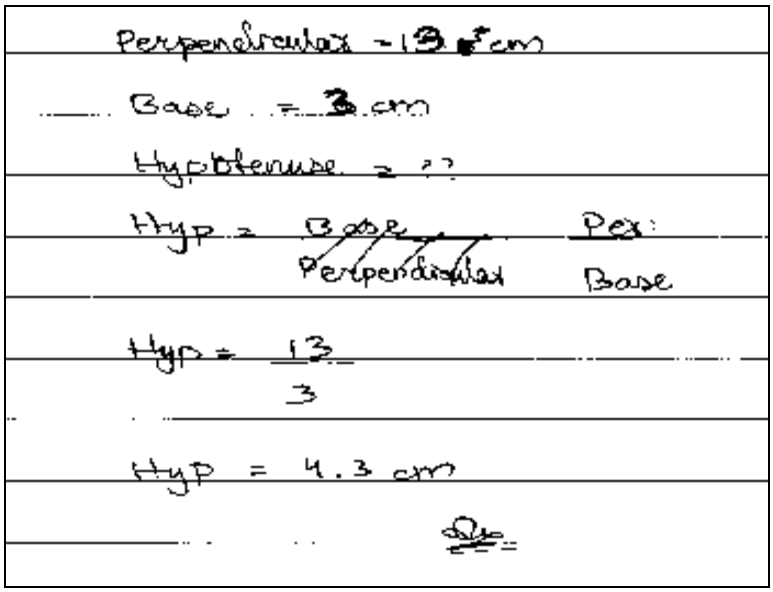
Better responses exhibited that candidates comprehended the question well. They correctly applied the Pythagorean Theorem in the triangle *ABC* to find the length of the side *AB*, which is common side of the triangle *ABD* and *ABC*. Finally, they found the length of *AD* using Pythagorean Theorem in the triangle *ABD*.

Example:

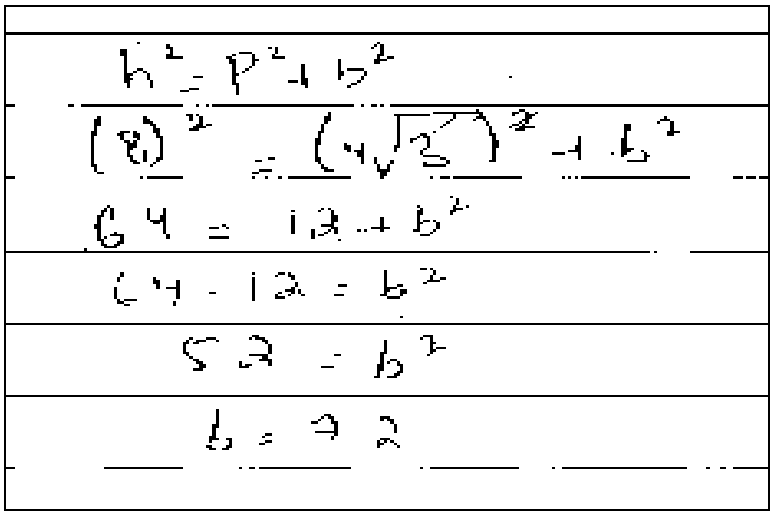


Weaker responses indicated that candidates failed to understand the question which led to wrong application of Pythagorean Theorem. Consequently, such responses failed to find the required length of *AD*. They made mistakes in application of theorem, identification of hypotenuse and other sides of the given triangles. The weaker responses also made mistakes in calculations as evident in the following examples.

Example 1:



Example 2:



Example 3:

From ΔABC,

Using pythagor's theorem,

$|\overline{AB}|^2 = |\overline{BC}|^2 + |\overline{AC}|^2$

$|\overline{AB}|^2 = (4\sqrt{3})^2 + (8)^2$

$|\overline{AB}|^2 = (4\sqrt{3})^2 + 64$

$|\overline{AB}|^2 = 12 + 64$

$\overline{AB} = \sqrt{76}$ cm

From ΔABD,

Using Pythagor's theorem,

$|\overline{AD}|^2 = |\overline{AB}|^2 + |\overline{DB}|^2$

$|\overline{AD}|^2 = (\sqrt{76})^2 + (3)^2$

$|\overline{AD}|^2 = 76 + 9$

$|\overline{AD}|^2 = 85$

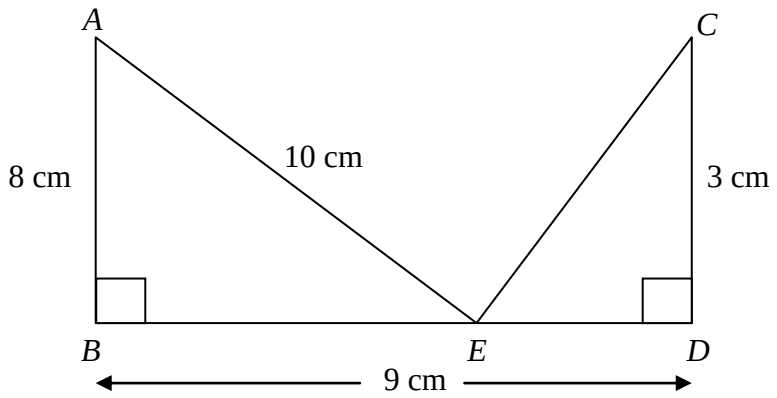
$\overline{AD} = \sqrt{85}$ cm

or 9.2cm

Question 8b:

In the given diagram, *ABE* and *CDE* are two right angled triangles. Calculate the length of *CE*.

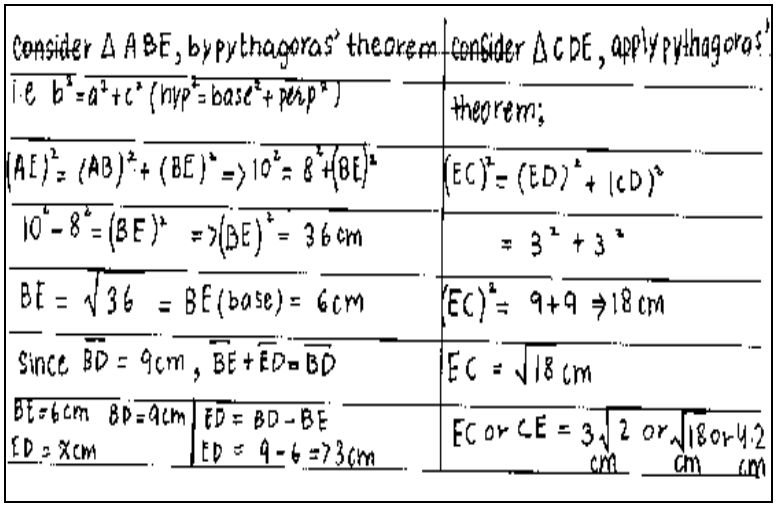
NOT TO SCALE



As compared to part **a**, this part was attempted by fewer candidates.

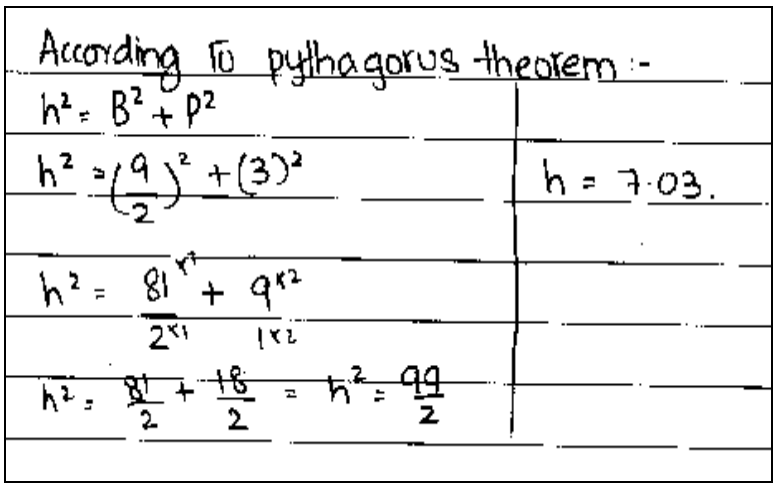
Better responses indicated that candidates understood the question well and were able to find the required lengths using Pythagorean's theorem correctly. Hence, they were able to find the length of side *ED* of triangle *CDE*.

Example 1:

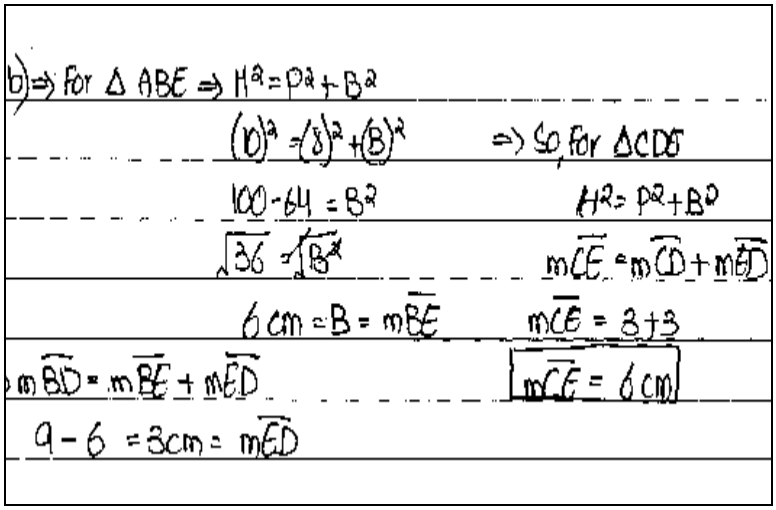


Weaker responses showed that candidates were unable to comprehend the given question and made different types of mistakes as cited in the following examples:

Example 1:



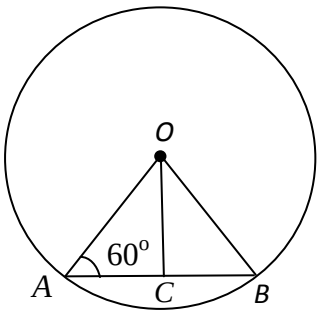
Example 2:



Question 9:

The given figure shows a circle having centre *O*. If *AC* = *BC*, then find $\angle AOC$ and $\angle AOB$. Also justify your answers.

NOT TO SCALE



This question was based on the theorem related to the circle and its properties.

Better responses showed that candidates had command over the theorems and their application. Candidates were able to find the required measurement of angles $\angle AOC$ and $\angle AOB$. They also were able to justify their answers with proper reason.

Example 1:

$\angle AOC = \angle COB = x$ ∵ angles opposite to congurrent sides are congurrent.	
Consider Δ AOC,	Consider Δ AOB,
$90^\circ + 60^\circ + x = 180^\circ$	$\angle AOB = 180^\circ - (60^\circ + 2x)$
$150^\circ + x = 180^\circ$	$= 180^\circ - (60^\circ + 2(30^\circ))$
$x = 180^\circ - 150^\circ$	$= 180^\circ - (60^\circ + 60^\circ)$
$x = 30^\circ$	$= 180^\circ - 120^\circ$
$\angle AOC = 30^\circ = \angle COB$	$\angle AOB = 60^\circ$

Example 2:

$m\angle ACO = m\angle BCO = 90^\circ$

(OC is perpendicular to AB so it bisect AB into two)

→ $m\angle AOC$

$m\angle AOC = 180^\circ - 60^\circ - 90^\circ$

$m\angle AOC = 30^\circ$ (All angles in a triangle is equal to 180°)

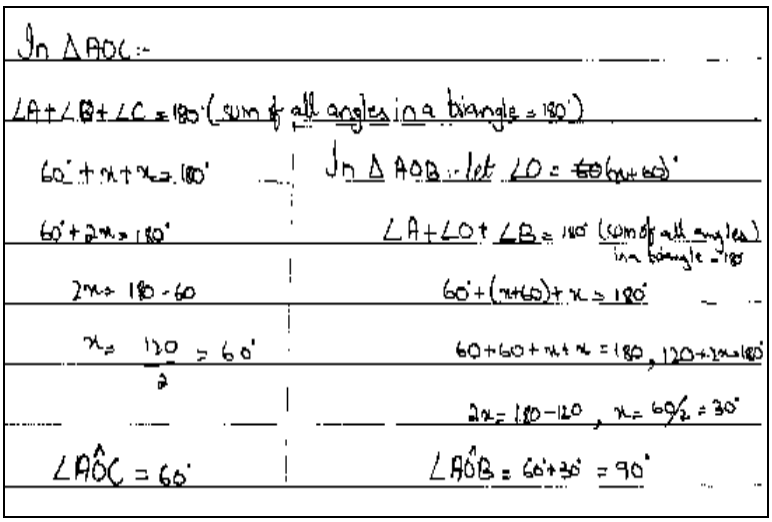
→ $m\angle AOB$

$m\angle AOB = m\angle AOC + m\angle BOC$

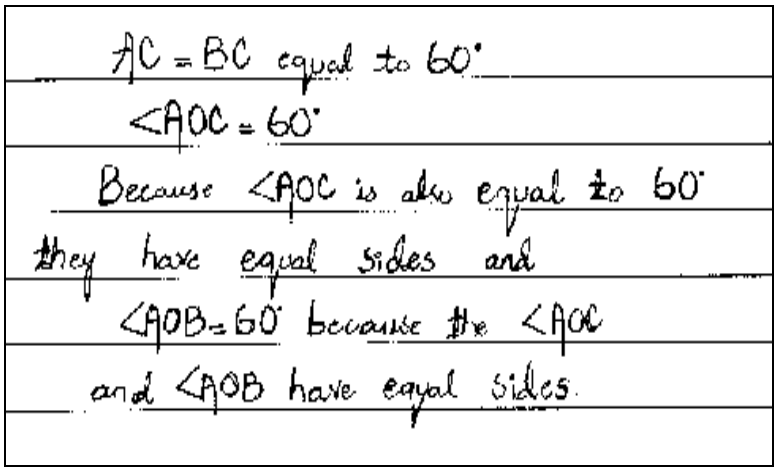
$m\angle AOB = 60^\circ$ (All angles in a triangle is equal to 180°)

Weaker responses indicated that the candidates failed to comprehend the properties of circles and therefore, were unable to find the required measurement or failed to justify their answers. The examples cited below exhibiting few misconceptions in weaker responses.

Example 1:

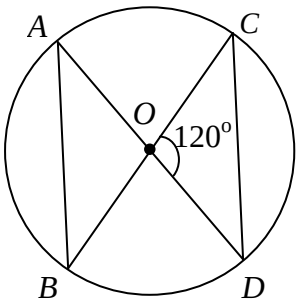


Example 2:



Question 10:

The given figure shows a circle having centre *O*. If *AB* = *CD*, then find



NOT TO SCALE

a. $\angle AOB$

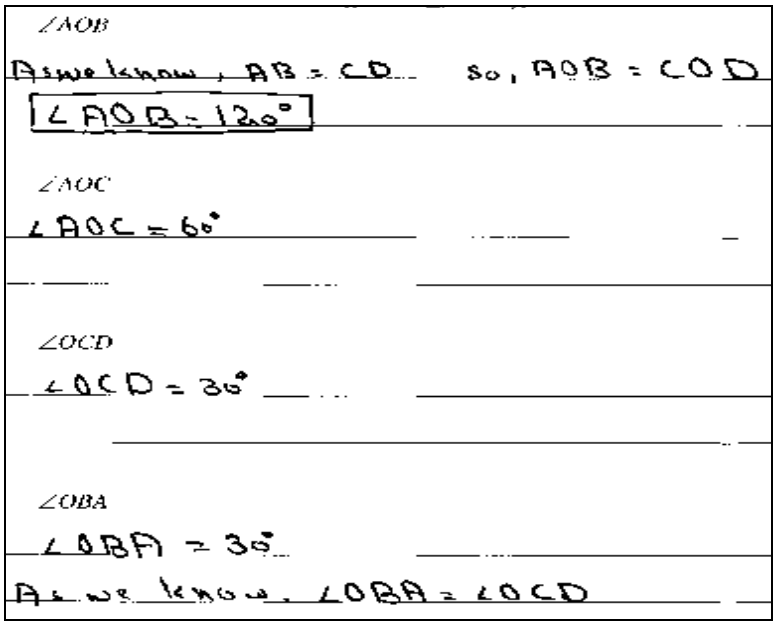
b. $\angle AOC$

c. $\angle OCD$

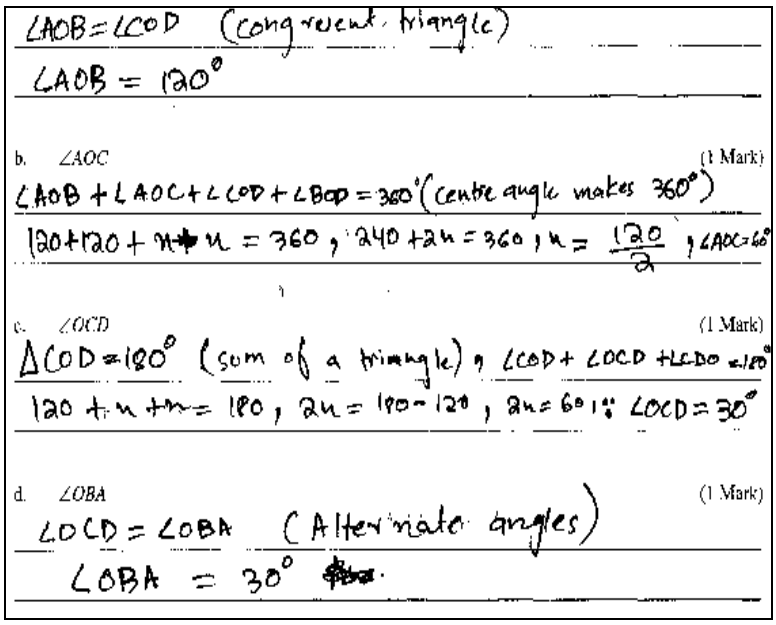
d. $\angle OBA$

Better responses displayed that candidates had command over the theorems and applied relevant theorems to find $\angle AOB$, $\angle AOC$, $\angle OCD$ and $\angle OBA$ correctly. The justification was not required in this question.

Example 1:

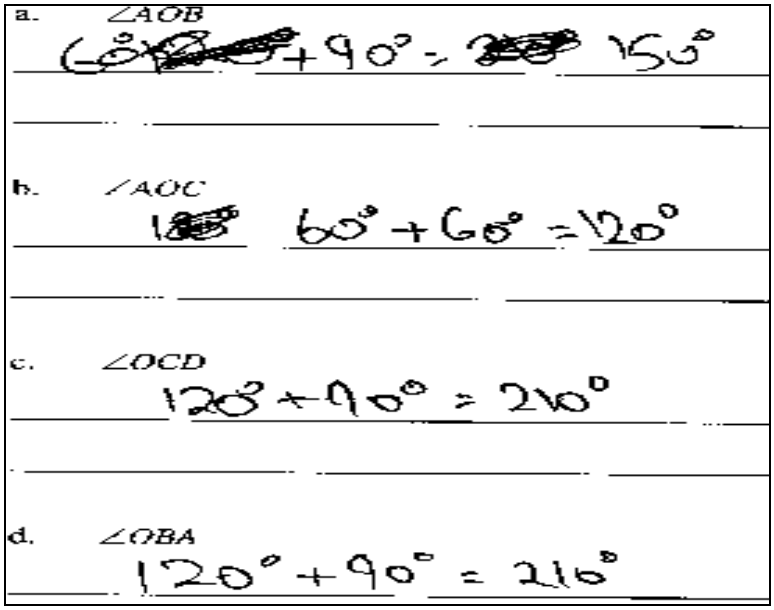


Example 2:

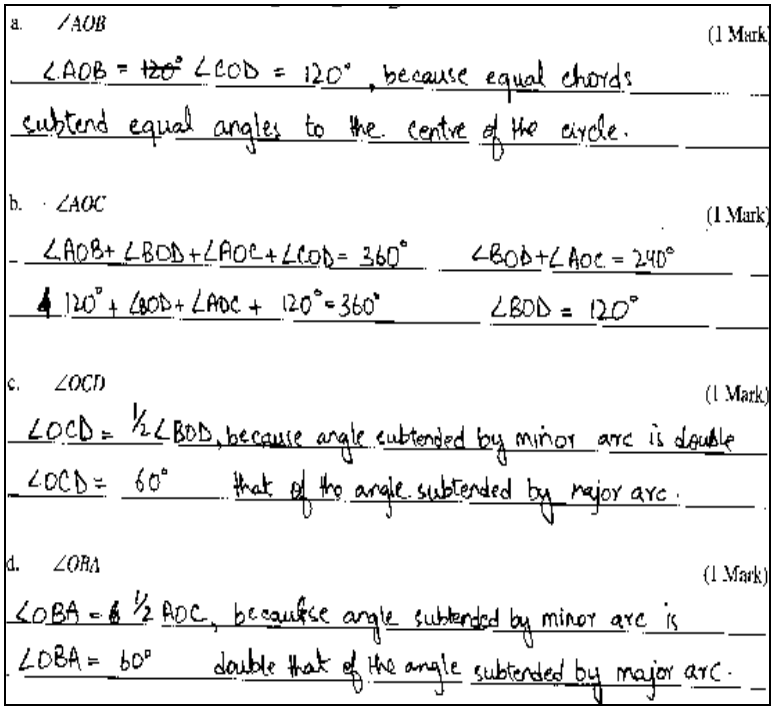


Weaker responses indicated that the candidates were unable to comprehend the theorem to be applied in the given question. Hence, they failed to fulfill the requirement of the question and found it difficult to find the required angle.

Example 1:



Example 2



a. ∠AOB (1 Mark)

∠AOB = ~~120°~~ ∠COD = 120°, because equal chords subtend equal angles to the centre of the circle.

b. ∠AOC (1 Mark)

∠AOB + ∠BOD + ∠AOC + ∠COD = 360° ∠BOD + ∠AOC = 240°

120° + ∠BOD + ∠AOC + 120° = 360° ∠BOD = 120°

c. ∠OCD (1 Mark)

∠OCD = ½∠BOD, because angle subtended by minor arc is double

∠OCD = 60° that of the angle subtended by major arc.

d. ∠OBA (1 Mark)

∠OBA = ½ AOC, because angle subtended by minor arc is

∠OBA = 60° double that of the angle subtended by major arc.

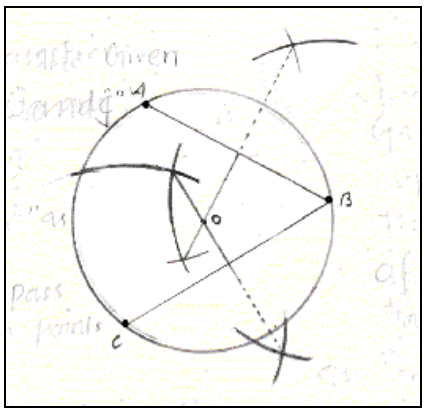
Question 11:

Draw a circle passing through the given three points in the given space. All necessary steps should be demonstrated.

This question was based on the construction of circle passing through the given three points.

Better responses displayed that candidates had good command over geometrical construction and had clear understanding of the process of drawing required circle. They first joined the given points and have drawn the right bisectors to get the centre of the circle and finally, drew the required circle.

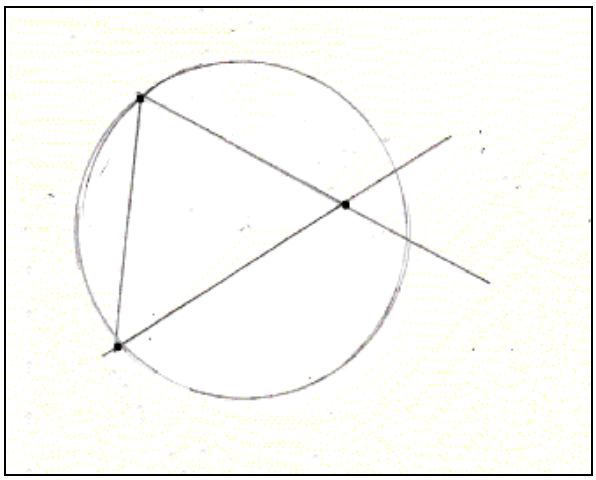
Example:



Weaker responses indicated that candidates had lack of understanding and failed to draw the required circle. Few have drawn the circle by using their judgment instead of following proper steps required to draw the circle. In few other responses it is noted that candidates have drawn angle bisectors instead of right bisectors of the sides hence failed to get the centre of the circle and consequently failed to draw the circle.

Two examples a cited below:

Example 1:



Example 2:

